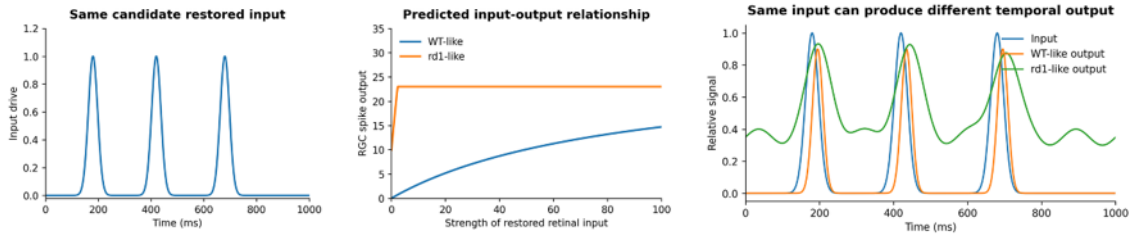


D Preliminary modelling concept: how altered electrophysiology could reshape retinal output

The same restored retinal input may be transformed differently if degeneration changes retinal ganglion cell physiology.



Illustrative test input

A common candidate restored signal is applied to both model states so that differences in output reflect the cell model, not the stimulus.

Conceptual output distortion

Degeneration could increase background firing, reduce usable dynamic range or cause output to saturate at lower input strengths.

Temporal coding

The same transient input could produce a cleaner, stimulus-locked response in WT-like cells and a broader, less distinct response in rd1-like cells.

How the grant will convert this concept into a biologically constrained model

Current clamp

Spontaneous firing

Step responses

Input-output curves

Adaptation

▶

Synaptic isolation

Separate intrinsic from network-driven abnormal activity

▶

Voltage clamp

Current amplitudes

I-V relations

Activation / inactivation kinetics

▶

Refined model

Replace illustrative parameters with measured WT and rd1 properties

▶

Functional readout

Which changes alter baseline firing, gain, timing and signal/background?

This panel does not claim what the animal sees. It illustrates how degeneration-associated physiology could change the retinal spike output sent to the brain.

Grant objective: identify which measured changes in surviving RGCs most strongly distort the transformation of restored input into retinal output.